

CUSTOMER BEHAVIOR ANALYSIS REPORT

Alfido Tech Internship

Submitted By

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Submission Date: 06-07-2026

1. Objective

The main objective of this project was to understand customer purchasing behaviour using an e-commerce dataset. I wanted to explore how customers differ based on their age, spending habits, payment methods, and purchasing patterns. Along with cleaning and analysing the data, I also performed customer segmentation using K-Means clustering to identify groups of customers with similar behaviour. The insights obtained from this analysis can help businesses understand their customers better and make informed marketing decisions.

2. Dataset Description

For this project, I used the **E-commerce Customer Behaviour Analysis** dataset available on Kaggle. The dataset contains approximately **250,000 customer transaction records**, making it suitable for performing large-scale customer behaviour analysis.

The dataset contains the following important attributes:

- Customer ID
- Purchase Date
- Product Category
- Product Price
- Quantity
- Total Purchase Amount
- Payment Method
- Customer Age
- Returns
- Customer Name
- Age
- Gender
- Churn

The information available in the dataset provides a complete overview of customer purchases, allowing different aspects of customer behavior to be explored.

3. Data Cleaning and Preprocessing

Before starting the analysis, I carefully examined the dataset to understand its structure and quality.

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked the number of rows and columns.
- Verified column names.
- Examined data types.
- Checked for missing values.
- Checked for duplicate records.
- Converted the Purchase Date column into datetime format.
- Prepared the dataset for further analysis.

Fortunately, the dataset was already well organised and required only a few preprocessing steps before analysis. Most of the work involved verifying data quality, converting the purchase date into datetime format, and preparing the dataset for further exploration.

4. Feature Engineering

To make the analysis more meaningful, a few additional features were created from the existing data.

The Purchase Date column was used to extract the Year and Month, making it easier to analyze purchasing trends over time.

An additional Age Group column was also created by dividing customers into Young, Adult, and Senior categories. This simplified customer comparison across different age groups and made the visualizations easier to interpret.

These newly created features improved both the analysis and the overall presentation of the results.

5. Exploratory Data Analysis

After cleaning the dataset, I explored it using summary statistics and different visualizations.

Statistical measures such as mean, median, mode, minimum, maximum, and standard deviation were calculated to understand the numerical variables.

The exploratory analysis focused on:

- Customer age
- Product categories
- Purchase amount
- Payment methods
- Customer churn
- Monthly purchasing trends

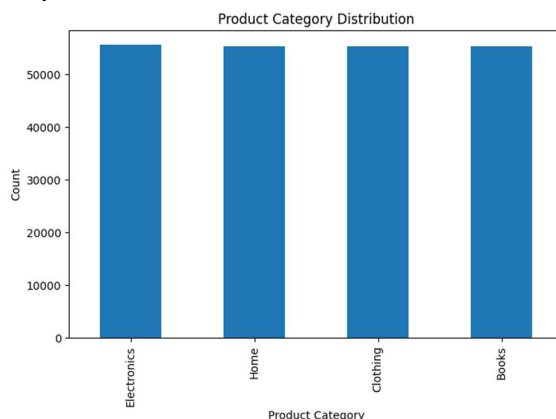
Exploring the dataset using summary statistics provided a better understanding of customer behaviour before creating visualizations. This made it easier to identify meaningful trends and patterns during the analysis.

6. Data Visualization

Several visualizations were created to better understand customer behaviour.

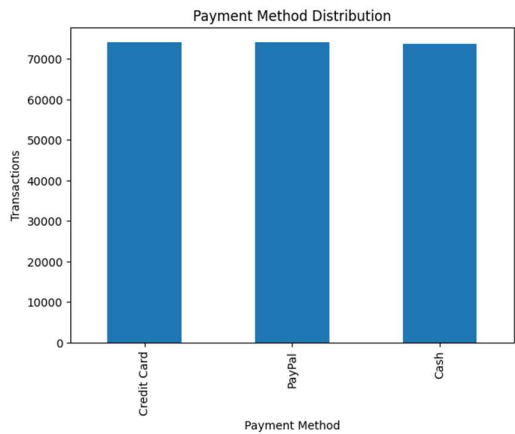
Product Category Distribution

A bar chart was created to compare the number of purchases across different product categories. This visualization helped me understand which product categories received more customer purchases. It also gave a quick overview of customer interest across different product categories.



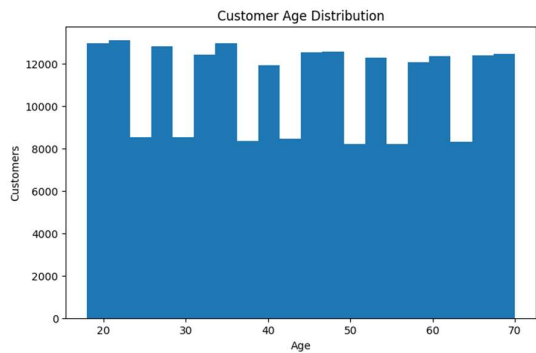
Payment Method Distribution

A second bar chart was created to analyze customer payment preferences. The chart shows that customers used multiple payment methods while completing their purchases.



Customer Age Distribution

A histogram was plotted to study the age distribution of customers. Most customers belonged to the adult age group, while fewer customers were found in the younger and senior age categories.



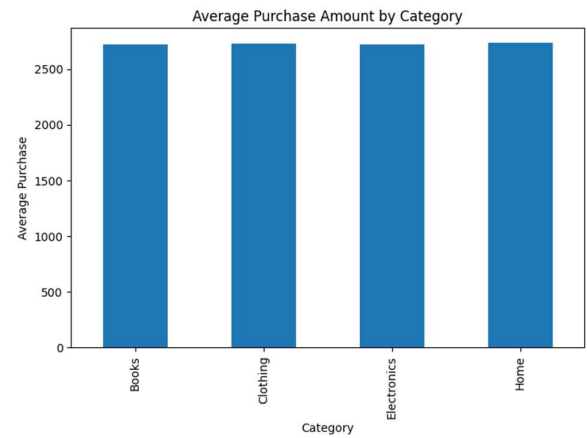
Monthly Sales Trend

A line chart was created to observe changes in total purchase amount over different months. This chart gives a simple overview of how customer purchases varied across different months.



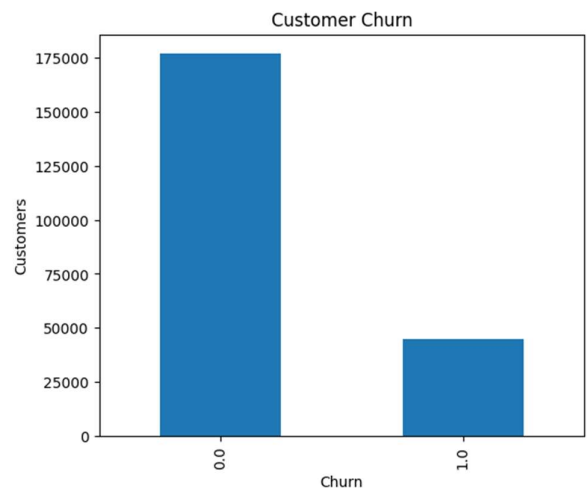
Average Purchase Amount by Product Category

This bar chart compares the average purchase amount across different product categories. It provides a better understanding of which product categories contribute more revenue per transaction.



Customer Churn Analysis

A bar chart was created to compare customers who continued purchasing with those who stopped purchasing. This analysis is useful for understanding customer retention.



7. Customer Segmentation using K-Means

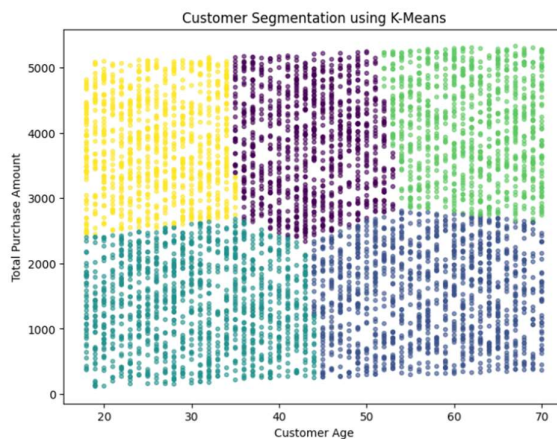
To understand customer purchasing behaviour more effectively, I applied the **K-Means Clustering** algorithm.

The clustering was performed using the following features:

- Customer Age
- Total Purchase Amount

After applying the K-Means algorithm, customers were grouped into five clusters using Customer Age and Total Purchase Amount. Each cluster represents customers with similar purchasing behaviour. This makes it easier for businesses to understand different customer groups and design more personalized marketing strategies.

The segmentation showed that customers have different spending patterns and purchasing characteristics. Instead of treating every customer the same, businesses can use these clusters to provide personalized recommendations, targeted promotions, and loyalty programs.



```
...      count
Cluster
1.0    45293
2.0    44760
3.0    44436
4.0    43982
0.0    43193
dtype: int64
```

8. Key Findings

After analysing the dataset and the generated visualizations, I observed several interesting patterns that helped me better understand customer purchasing behaviour.

- The dataset contained approximately 250,000 customer purchase records suitable for business analysis.
- Customer purchases were distributed across multiple product categories.
- Customers used different payment methods rather than relying on a single payment option.
- Purchase amounts varied significantly among customers.
- Monthly sales showed changes throughout the year, indicating variations in purchasing activity.
- Customer churn analysis highlighted the importance of customer retention.
- K-Means clustering successfully grouped customers into five different segments based on customer age and purchase amount.

9. Recommendations

Based on the analysis, I suggest the following recommendations:

- Develop personalized marketing campaigns for different customer segments.
- Reward high-value customers through loyalty programs and exclusive offers.
- Provide attractive discounts to customers who show signs of churn.
- Focus promotional activities on product categories with higher average purchase amounts.
- Analyze monthly sales trends before planning seasonal campaigns.
- Continue monitoring customer behaviour regularly to improve business decisions

10. Conclusion

Working on this project gave me practical experience in analysing a real-world e-commerce dataset using Python. Throughout the project, I learned how to clean data, perform exploratory data analysis, create visualizations, and apply K-Means clustering for customer segmentation. The project also helped me understand how raw customer data can be transformed into meaningful business insights that support better decision-making. Overall, this project improved both my technical skills and my confidence in working with real business datasets.
